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Perceptually Equivalent Judgments Made Visually and via Haptic Sensory-Substitution Devices

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ABSTRACT

According to the ecological theory of perception–action, perception is primarily of affordances, which are directly perceivable opportunities for behavior. The current study evaluated participants' ability to use vision and haptic sensory-substitution devices to support perceptual judgments of affordances involving the task of passing through apertures. Sighted participants made perceptual judgments about whether they could walk through apertures of various widths and their level of confidence in each judgment, using unrestricted vision and, when blindfolded, using two haptic sensory-substitution instruments: a cane-like wooden rod and the Enactive Torch, a device that converts distance information into vibrotactile stimuli. The boundary between aperture widths that were judged as pass-through-able versus non-pass-through-able was statistically equivalent across sensory modalities. However, participants were not as confident in their judgments using the rod or Enactive Torch as they were using vision. Additionally, participants' judgments with the haptic instruments were significantly more accurate than with vision. The results underscore the need to assess sensory-substitution devices in the context of functional behaviors.

To say that a behavior is afforded is to say that the behavior is possible to perform given the layout of environmental surfaces and substances taken in reference to an individual (Stoffregen, 2003; see also Chemero, 2009; Gibson, 1966/1983, Gibson, 1966/1983, 1979/1986; Greeno, 1994; Jones, 2003; Reed, 1996; Turvey, 1992; Warren, 1984). The behavior of typical walking (i.e., without rotating the shoulders) through an aperture, for example, is afforded when the width of the aperture is greater than or equal to a certain proportion of the width of a person's shoulders. Warren and Whang (1987) demonstrated that sighted perceivers are visually sensitive to the boundary between pass-through-able and non-pass-through-able. Their findings are consistent with many other studies demonstrating that sighted participants are able to visually determine whether many activities are afforded (see review by Fajen, Riley, & Turvey, 2009), such as the catchableness of moving objects

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(Oudejans, Michaels, Bakker, & Dolne, 1996), the crossableness of gaps (Burton, 1992, 1994), sit-on-ableness (Mark, 1987; Mark, Balliet, Craver, Douglas, & Fox, 1990), and step-on-ableness (Ramenzoni & Riley, 2005; Warren, 1984), to name just a few.

The ability to perceive affordances haptically is also well documented. For example, people can haptically perceive affordances of hand-held objects, such as whether tools and implements are hammer-with-able and poke-with-able (Wagman & Carello, 2001; Wagman & Shockley, 2011; see also Hove, Riley, & Shockley, 2006), move-able (Shockley, 2009; Shockley, Carello, & Turvey, 2004), and reach-with-able (Carello, Thuot, Anderson, & Turvey, 1999). As is apparent from the ability of visual impaired cane users to effectively move about, people can also use haptics to probe the environment with regard to affordances related to mobility. Sighted individuals can use a cane to perceive affordances for the crossableness of gaps in the locomotor substrate (Burton, 1992, 1994; Burton & Cyr, 2004) and the stand-on-ableness of slopes (Fitzpatrick, Carello, Schmidt, & Corey, 1994), for example.

The perceptual ability to perceive affordances accurately, regardless of modality, is adaptive in that it permits *prospective* control of behavior (Turvey, 1992). For example, prior to attempting to pass through an aperture, sighted individuals can visually distinguish those that are wide enough to afford passage from those that are not (e.g., Davis, Riley, Shockley, & Cummins-Sebree, 2010; Fath & Fajen, 2011; Franchak, Celano, & Adolph, 2012; Higuchi, Cinelli, Greig, & Patla, 2006; Wagman & Taylor, 2005; Warren & Whang, 1987). Prospective control based on affordance perception may permit behavioral strategies that are safer and more efficient than trial-and-error learning.

The preceding concepts, which are central to the ecological approach to perception–action, may have practical importance by informing the development of sensory-substitution devices. A sensory-substitution device is a tool that allows the user to do with one sense what is typically done with another (Auvray & Myin, 2009; Bach-y-Rita, Collins, Saunders, White, & Scadden, 1969; Bach-y-Rita & Kercel, 2003; Lenay, Gapenne, Hanneton, Marque, & Genouelle, 2003; Maidenbaum, Abboud, & Amedi, 2014; Segond & Weiss, 2005). The purpose of sensory-substitution devices is to enable functional activity by informing the user about the world (and the behaviors it affords the user) using a sensory modality (e.g., haptics) that is available to the user when another modality (e.g., vision) is unavailable or compromised, for example, due to a visual deficit. The challenge regarding their development is how to effectively relay information typically detected through one modality (e.g., information about ego-centric distance to an object, which might typically be detected visually) through the substituting modality (e.g., haptically via vibrations of varying intensity applied to the skin).

Lenay et al. (2003) provide a useful description of sensory-substitution systems as generally comprising three main parts. *Sensor(s)* respond to some form of energy, such as light in the infrared spectrum, to obtain a description of the environment. A *coupling system*, typically instantiated in a computational platform such as Arduino hardware and software platforms, “interprets” the detected stimulus energy. A *stimulator* then outputs the interpreted stimulus energy into another form, such as haptic vibrations. Accordingly, a sensory-substitution device that converts light into vibrations would be a visual–tactile device.

One goal of sensory-substitution devices is to facilitate the ability to carry out goal-directed actions. In keeping with the goal for sensory-substitution devices to enable functional activity, it is desirable for those devices to permit safe and efficient prospective control strategies for their users. People encounter apertures frequently in the everyday environment, including fixed apertures such as doorways and dynamic ones such as gaps between

other people in a crowded plaza. Thus, prospective judgments of whether apertures afford passage present a behaviorally relevant task for assessing the suitability of sensory-substitution devices for enabling functional mobility. However, research on sensory-substitution devices has yet to determine whether a visual–tactile device can support accurate perceptual determination of the boundary between passable and non-passable apertures.

Visual judgment of aperture pass-through-ability has been studied in a variety of ways. Warren and Whang (1987) studied the ability to detect *critical points*—aperture widths that separate different behavioral modes—of visually guided walking through apertures. They identified the actual critical aperture-to-shoulder-width ratio (A/S , where A is width of the aperture and S is shoulder width at the broadest point) separating the transition from passing through the aperture while walking normally to needing to rotate the shoulders to walk through to be $A/S = 1.30$. After determining that critical point, Warren and Whang had observers with small and large S provide prospective, visual pass-through-ability judgments (i.e., a yes/no response when presented with an aperture of a certain width) in a static condition (participants stood at a fixed point of observation) and in a moving condition (participants walked in a straight line toward the front of the aperture before giving reports). Judgments of the critical A , when expressed relative to S , were consistent across all conditions, ranging from $A/S = 1.14$ to 1.17 . These results are in accordance with Warren and Whang's hypotheses that participants judge category boundaries between pass-through-able and not-pass-through-able apertures as a constant ratio of their body size and that static monocular information is sufficient for perceiving the affordance of pass-through-ability.

Other studies have confirmed and expanded on Warren and Whang's (1987) finding that people can accurately perceive the affordance of pass-through-ability. Higuchi et al. (2006) demonstrated that while approaching apertures of varying widths, participants who held horizontal bars that increased their effective body width rotated their shoulders to degrees proportional to the width added to their body by the bar. Wagman and Taylor (2005) similarly found that participants' visual judgments of pass-through-ability when holding a wide implement in the hand accurately took into account information about body-plus-implement width when information about the latter was detected haptically. Gordon and Rosenblum (2004) found that participants could make auditory judgments of pass-through-ability on the basis of sounds emanating from a source on the other side of an aperture. In many of these studies, participants utilize static, eye-height-scaled optical information. Fath and Fajen (2011; see also Fajen, Diaz, & Cramer, 2011) contributed to this area of research by investigating the ability of participants to use multiple kinds of dynamic informational variables. Informational variables are “dynamic” when they are only available to perceivers as they move (Fath & Fajen, 2011) and generate patterns of optical flow, which, in the case of forward motion, is the radial expansion of light in the field of vision corresponding to the relative movement of observer and environment (Favela & Chemero, 2016; Gibson, 1979/1986). Fath and Fajen (2011) demonstrated that dynamic informational variables in optic flow arising from walking could be utilized by participants to inform perceptually guided locomotion through apertures, thereby reinforcing the dynamic nature of perception.

Many other studies in addition to these have also investigated locomotion, oculomotor behavior, and other sensorimotor processes related to aperture passage (e.g., Baker & Cinelli, 2017; Davis et al., 2010; Franchak et al., 2012; Fujikake, Higuchi, Imanaka, & Maloney, 2011; Hackney, Cinelli, Denomme, & Frank, 2015; Hackney, Cinelli, & Frank, 2014; Higuchi, Cinelli, & Patla, 2009; Higuchi et al., 2011; Keizer, De Bruijn, Smeets, Dijkerman, & Postma,

2013; Wilmut & Barnett, 2011). Although there is varied research on affordance perception involving apertures, there is far less research utilizing sensory-substitution devices to detect affordance boundaries (e.g., Fitzpatrick et al., 1994; Wagman & Hajnal, 2016).

In the current study, we assessed judgments of the pass-through-ability of apertures using a haptic sensory-substitution device, the Enactive Torch (Figure 1; Froese, McGann, Bigge, Spiers, & Seth, 2012; Grespan et al., 2008), in comparison to visual judgments and haptic judgments made with the assistance of a wooden rod used to probe the environment. The Enactive Torch is a sensory-substitution device that detects distance using an infrared range finder and converts distances into vibrotactile stimuli applied to the wrist. Previous studies investigating the efficacy and utility of the Enactive Torch have focused on navigation tasks (Froese et al., 2012) and did not determine whether the Enactive Torch supports perception of affordance boundaries. However, prior research suggests that accurate affordance

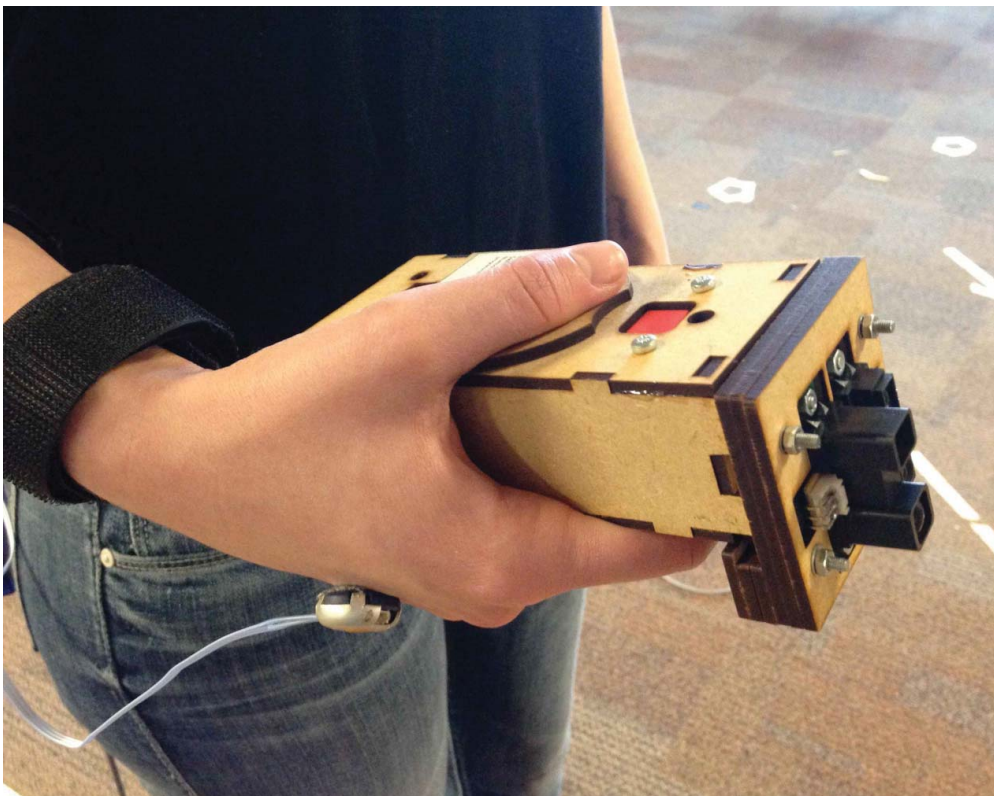


Figure 1. The Enactive Torch, version 5. This model was utilized in the experiment. The infrared range sensors are at the front. The device connects via a cord to the vibrational motor, which is attached by a Velcro strap to the user's wrist. Vibration intensity is inversely proportional to the distance of a surface that is currently detected by one of two range sensors: A smaller range sensor detects distances of 8–80 cm and a larger sensor detects distances of 20–150 cm. The device runs software on an Arduino Pro-mini 5V. The following data-processing capabilities of the device were utilized: distance (*Sensor1* and *Sensor2*) and haptic vibration output (*MotorValue*). Although not utilized in the current experiment, the device can also process movement (triple-axis accelerometer: *AccX*, *AccY*, *AccZ*) and audio output (*ToneFrequency*). The Enactive Torch was designed by Tom Froese and Adam Spiers (2007). To learn more visit <https://enactivetorch.wordpress.com>.

perception could be supported by another visual–tactile sensory-substitution device. Travieso, Gomez-Jordana, Diaz, Lobo, and Jacobs (2015) examined the use of a vibrotactile sensory-substitution device to support perception of the climbability of steps. The device they investigated, like the Enactive Torch, conveyed vibration at intensities that varied as a function of object-to-sensor distance. However, unlike the Enactive Torch, which is hand wielded and conveys vibrotactile stimulation to a single point on the wrist, their device utilized 24 coin motors, positioned in a vertical strip worn on the torso (see Diaz, Barrientos, Jacobs, & Travieso, 2012). The purpose of that arrangement was to generate “vibrotactile flow” (Cancar, Diaz, Barrientos, Travieso, & Jacobs, 2013; Diaz et al., 2012), which, like optical flow (Gibson, 1979/1986; Lee, 1976), was hypothesized to specify affordances such as step-on-able (Travieso et al., 2015) and to convey the presence of ground-level obstacles (Diaz et al., 2012). Their results largely echoed the results typically obtained using vision, indicating that their vibrotactile device supported the detection of critical points separating climbable from non-climbable steps.

Kolarik, Timmis, Cirstea, and Pardhan (2014) similarly determined whether sensory-substitution devices support perception of mobility-related affordances. Blindfolded participants in their study walked through apertures using echo-based sensory-substitution devices. Participants were able to use information from the devices to modulate shoulder rotations, although they made larger modulations, exhibited greater movement times, walked more slowly, and occasionally collided with the aperture boundaries. Kolarik et al. did not assess prospective judgments of whether participants believed they could pass through the apertures.

The ubiquity of aperture passage in daily life and the abundance of experimental data on perceptual judgments of this ability make this task well suited for assessing the utility of visual–tactile sensory-substitution devices such as the Enactive Torch for supporting functional mobility (see also Kolarik et al., 2014). The current study assessed the relative accuracy and confidence of sighted participants’ prospective judgments of their ability to walk through an aperture without turning the shoulders. Judgments were made in three conditions: using vision, using haptic perception by physically probing the aperture with a cane-like wooden rod (cf. Barac-Cikoja & Turvey, 1991, 1993), and using vibrotactile haptic perception with the Enactive Torch. The perceived critical boundaries separating pass-through-able from non-pass-through-able apertures were compared across those modalities and, within each modality, to each participant’s actual aperture pass-through-ability (without turning their shoulders).

In order to perceive whether or not an aperture affords passage, a person needs to detect information about the relation between the self (i.e., shoulder width) and environment (i.e., aperture width). This perceptual task is thus fundamentally a case of *exproprioception*, that is, perception of the body in relation to the environment (Baker & Cinelli, 2017; Pagano, Carello, & Turvey, 1996). It is possible that eye-height-scaled optical information (Warren & Whang, 1987) or dynamic optical variables such as aperture width scaled by head sway or stride length to shoulder width (Fath & Fajen, 2011) can serve an exproprioceptive function with regard to detecting aperture pass-through-ability. Another possibility is that aperture pass-through-ability perception requires sensitivity to higher order variables in the global array (Stoffregen & Bardy, 2001) comprised of relations between lower order optical and mechanical (i.e., proprioceptive) variables. In either case, in the vision condition of the present experiment information about pass-through-ability could arise from head movements and postural

sway, which would create concurrent fluxes in the optical and mechanical arrays that specify aperture width relative to shoulder width (Chen & Stoffregen, 2012; Stoffregen, Chen, Varlet, Alcantara, & Bardy, 2013; Stoffregen, Pagulayan, Bardy, & Hettinger, 2000). In the two haptic conditions (i.e., wooden rod and Enactive Torch conditions), it seemed that there would be a way of similarly detecting angular aperture width relative to shoulder width. In the wooden rod condition, we expected the utilization of information variables via mechanical stimulation of mechanoreceptors resulting from the combined effects of the exploratory movements—primarily horizontal angular excursions—of the rod (Barac-Cikoja & Turvey, 1993; Hove et al., 2006) in relation to the rod's inertial properties (Burton, 1992; Wagman & Carello, 2001) and the mechanical energy arising from strikes of the rod against the aperture (Barac-Cikoja & Turvey, 1991). In the Enactive Torch condition, strikes against the aperture were absent, but exploratory movements (i.e., sweeping the Enactive Torch back and forth) in combination with vibrotactile stimulation at an intensity proportional to the distances detected by the infrared sensors served as the primary means of detecting information about aperture pass-through-ability.

In light of the existing research on aperture pass-through-ability and the use of sensory-substitution devices to support mobility-related affordance perception, our hypothesis was that vision would support more accurate affordance judgments than both the rod and the Enactive Torch. First, this claim was motivated by the large amount of experimental work on visual perception demonstrating that humans accurately perceive boundaries between conditions that afford an action and those that do not (for a very small sample see Fath & Fajen, 2011; Gibson, 1979/1986; Lee, 1976; Mark, 1987; Oudejans et al., 1996; Warren, 1984; Warren & Whang, 1987). Second, this claim was motivated by the work of Kolarik et al. (2014) and by other studies showing that, unlike visual performance, participants underestimate aperture width when using a probe while blindfolded (Barac-Cikoja & Turvey, 1991), and participants tend to be more conservative when making judgments about the crossability of gaps when using a probe while wearing a blindfold (Burton, 1992). We do not offer a specific hypothesis regarding perceptual performance with the Enactive Torch compared to the rod because there is no precedent for research involving use of the Enactive Torch to support affordance judgments. The work of Travieso et al. (2015) does, however, suggest that a similar vibrotactile device supported effective determination of step-on-ability, although, as discussed in the preceding, the perception of affordances via the Travieso et al. device was informed by patterns of change in vibrotactile flow specifying time to contact. The Enactive Torch is hypothesized to specify affordances via the pattern of intensity changes of vibrotactile stimulation applied to the user's arm during exploratory movements.

Method

Participants

Twenty-seven undergraduate students (19 women and 8 men) from the University of Cincinnati participated in this study. This sample size is based upon similar past experiments involving visual aperture pass-through-ability judgments (Warren & Whang, 1987) and aperture size judgments made with the assistance of a rod for striking (Barac-Cikoja & Turvey, 1991; Hanley & Goff, 1974). Participants' ages ranged from 18 to 42 years ($M = 22.31$,

$SD = 6.17$ years), and shoulder widths ranged from 37 to 50 cm ($M = 42.48$, $SD = 3.31$ cm). To maintain measurement consistency, shoulders were measured with a constructed vernier-like caliper from the heads of the left and right humeri, with measurements taken from the front of the participant (cf. Warren & Whang, 1987) and the back. All participants reported no history of movement disorders or experience using a cane or other handheld mobility assistance device. All participants reported normal or corrected-to-normal vision (with contact lenses or eyeglasses), the ability to walk without assistance, and no other sensory deficits.

Materials and apparatus

Consistent with previous experiments involving perceptual judgments of the pass-through-ability of apertures (Davis et al., 2010; Warren & Whang, 1987), a two-paneled, sliding



Figure 2. Aperture utilized during experiment. Aperture (height: 2.43 m; width: 2.6 m) with two sliding panels (height: 2.43 m; width: 40.5 cm, each). Panels slid laterally across the frame to create aperture widths ranging from 40 cm to 75 cm.

doorway attached to a platform was constructed (Figure 2). Two moveable panels (each 2.43 m tall, 40.5 cm wide) were inserted into a standing wooden frame (2.43 m tall, 2.6 m wide). The panels could slide laterally across the frame to create aperture widths ranging 40 to 75 cm. Two platforms (each 8.9 cm tall, 1.23 m long, and 0.91 m wide) were attached at the middle front and back of the frame to form a continuously level “walkway” that ran through the aperture, perpendicular to the panels.

Participants utilized sensory-substitution devices in two of the three conditions: a cane-like wooden rod and the Enactive Torch. The rod’s dimensions (length: 121.5 cm, diameter: 1.27 cm; weight: 113.4 g) matched specifications of those utilized in similar, previously published experiments (cf. Burton, 1992, 1994), which were themselves based on cane dimension recommendations set by the American Foundation for the Blind (Farmer, 1978).

The Enactive Torch (Froese et al., 2012; Grespan et al., 2008) is a flashlight-sized (length: 15.8 cm; width: 5.8 cm; height: 4.6 cm; weight: 350 g) sensory-substitution device that is held with one hand and connected by a wire to a wrist strap with a vibrational motor (Figure 1). The Enactive Torch uses infrared range sensors to detect distances to adjacent surfaces. Those distances are translated to haptic (vibrational) stimuli by a motor embedded in the wrist strap worn by the user. Vibration intensity is inversely proportional to the distance of a surface that is currently detected by one of two range sensors: A smaller range sensor detects distances of 8–80 cm and a larger sensor detects distances of 20–150 cm. Sensors sample distance at 100 Hz and the data from the two range sensors are combined via software running on an Arduino Pro-mini 5V. By sweeping the Enactive Torch back and forth the user obtains information about the layout of nearby surfaces (i.e., their relative distances) in terms of a time-varying pattern of vibrations of changing intensity.

Participants wore a blindfold in the rod and Enactive Torch conditions. In all three conditions, participants wore earmuffs (3M PELTOR Sport Bull’s Eye 9) with a noise reduction rating of 25 dB. Such a noise reduction rating is able to attenuate sounds as loud as a jet taking off at a distance of 305 m (cf. Berger, Royster, Royster, Driscoll, & Layne, 2003), which can also block participants from hearing the sounds of strikes against the aperture with the rod and sounds from the Enactive Torch’s vibrating motor on their wrist. It is important to block auditory information during the trials because such information can influence participants’ affordance judgments. For example, Gordon and Rosenblum (2004) demonstrated that with blocked vision, sighted participants were sensitive enough to use sound to make body-scaled assessments regarding the passability of apertures.

Procedure

Participants provided informed consent after hearing an overview of the experimental procedure and reading the institutional review board-approved study procedures and consent document. Once participants granted consent, their demographic information was recorded; this included sex, age, ethnicity/race, height, weight, and handedness. Participants were then screened for visual acuity by being placed at a distance of 6.1 m from a wall-mounted, standard Snellen chart and were asked to read the eighth row of letters. Participants unable to read this row were dismissed. Next, participants were screened for prior regular cane usage. “Prior regular cane usage” referred to usage over periods of time long enough for a participant to learn the ability to make accurate perceptual judgments. Although there is no general consensus on how long it takes an individual to learn to successfully navigate with a cane or cane-like instrument,

it is generally accepted that approximately 3 weeks of intensive training would be necessary (cf. Altman & Cutter, 2004; Bickford, 1993; Ludt & Goodrich, 2002). Thus, 3 weeks was the cutoff criterion. The screening procedures resulted in no participants being eliminated from the study, yielding the final sample size of 27 as reported in the preceding.

Participants who passed the screening procedures underwent a familiarization and training task. The goal of this was to instruct participants to make rudimentary judgments with the rod and Enactive Torch, and to familiarize them with the operation of the latter. Participants were blindfolded and wore earmuffs. They were then placed 76 cm from one of the outside edges of the frame that was used in the actual experiment to create the apertures. Participants were asked to use the rod and Enactive Torch to make contact (or virtual contact in the case of the Enactive Torch) with the outside of one side of the aperture frame and then walk forward until they believed their body was lined up with the frame, specifically, that the frame was parallel to the shoulder of the hand with which they were using the tool. This was done three times with each tool and took approximately 10 minutes. Participants were not presented with any actual apertures at this time. All participants were able to perform the familiarization and training task.

Next, participants were placed in front of the aperture. Participants stood on the platform with the heels of their feet 76 cm from the aperture. Participants were told that they would remain at this same distance from the aperture for the entire experiment. This distance was standard across all participants and was based on the minimum distance at which the rod could be utilized to explore the openings of the widest presented aperture. Participants were presented each of the three modality conditions in blocks, and blocks were presented in random order across participants. The three modality conditions were (a) vision; (b) blindfolded with rod; and (c) blindfolded with Enactive Torch. For the vision condition, participants had their head centered with respect to the aperture, and for the two haptic conditions participants had the shoulder of the arm utilizing the tool centered with the aperture, so that across conditions the anatomical site of information detection was held constant with respect to the edges of the aperture such that exploratory movements to the left and right would offer similar information about aperture width.

For each condition participants were presented 16 aperture widths that ranged in 5-cm increments from 40 to 75 cm. Each of the eight widths was randomly presented once and then randomly presented again, for a total of 16 trials per condition block. Participants were not informed about the range of widths or that any of the widths would be repeated. Participants were asked to provide perceptual reports concerning their ability to pass through the aperture by responding “yes” or “no” with regard to whether they could comfortably walk through the presented aperture on a given trial, where “comfortably” was defined for them with regard to walking without rotating the shoulders and without hitting the panels with their arms. After providing their “yes” or “no” response, participants were asked to provide a confidence rating between 1 and 7 (1: *not confident in my judgment*; 7: *very confident in my judgment*). Participants were not provided any instructions concerning how to explore the environment with either the rod or the Enactive Torch. Participants were asked to close their eyes between trials in the vision condition, and they were asked to point the rod and Enactive Torch to the side between trials in those respective conditions while the panels were moved to create the aperture width for the next trial. Participants were asked to sit down after each block.

Once all trials were completed, measurements were taken of the participants’ shoulder width and actual minimum passable aperture. In order to obtain the latter, participants

began at the edge of one platform, walked through to the edge of the other platform on the other side of the aperture (for a total distance of 2.54 m), and then turned and walked back. This was done multiple times until an aperture opening was found that was minimally wide enough to afford passage for the participant without collision or shoulder rotation.

Results

The actual critical aperture-to-shoulder-width ratio at normal walking speed for all participants was $A/S = 1.36$ ($SD = 0.11$). The point of subjective equality (PSE)—the aperture width that elicited 50% “yes” and 50% “no” responses—served as the category boundary separating apertures perceived as pass-through-able from those that were not (Avraham et al., 2012; Engen, 1971; Meese, 1995). The PSE was estimated by means of a logistic regression function that related the percentage of “yes, pass-through-able” responses to aperture sizes scaled by the actual minimum aperture width each participant could pass through, with a PSE closer to 1.0 indicating greater accuracy (Figure 3A). A one-way, within-subjects analysis of variance (ANOVA) was performed to compare the PSE values obtained from judgments of passability made with vision, rod, and Enactive Torch. The main effect of modality was marginal but not significant, $F(2, 26) = 2.91$, $p = .06$, $\eta_p^2 = 0.1$. A post hoc power analysis, fitted to the parameters of the experiment, demonstrated that a sample size of 27 attained a power of .98 with an effect size of $\eta_p^2 = .10$.

A two one-sided test (TOST) for equivalence was used to verify the absence of a modality effect on PSE. Using a confidence interval of 90%, average PSE across vision, rod, and Enactive Torch each fell within the region of scientific indifference within the other modalities, as defined by the minimum and maximum PSE for each condition, $p < .001$ (Lakens, 2017; Wagman & Hajnal, 2014; Walker & Nowacki, 2011). Additional tests comparing the mean of each modality separately to the hypothetical value of 1 (i.e., accurate judgment) were conducted. One-tailed t -tests with Bonferroni corrections showed that judgments made with vision were significantly lower than accurate judgments (i.e., than 1), $t(26) = -5.37$, $p < .001$, whereas judgments made with the rod [$t(26) = -1.44$, $p = .081$] and Enactive Torch [$t(26) = -1.84$, $p = .039$] did not significantly differ from accurate judgments. It is important to highlight that the “mean of each modality” included here refers to the mean PSE that was obtained, as stated earlier, via a logistic regression function relating the percentage of “yes, pass-through-able” responses to aperture sizes already scaled by the actual minimum aperture width each participant could pass through. Consequently, the accurate judgment of 1 is an action-scaled and not absolute value.

A univariate ANOVA was used to examine the relationship between modality during the first trial block for each participant and the corresponding PSE during the first block in order to check for order effects and found no significant effect of order of presented modality, $p > .05$. A repeated-measures ANOVA examined whether PSE changed over time (i.e., from block 1, block 2, and block 3) independent of modality. Results indicated that PSE was statistically equivalent across presentation blocks, counterbalanced by modality, $p > .05$.

Participants’ confidence ratings (1 indicating low confidence in judgment and 7 indicating high confidence in judgment; Figure 3B) ranged from 2.76 to 6.77 for vision ($M = 4.4$, $SD = 1.2$), from 2.56 to 4.89 for the rod ($M = 3.86$, $SD = 0.71$), and from 1.34 to 6.45 for the Enactive Torch ($M = 3.82$, $SD = 1.24$). These confidence ratings were also analyzed using a one-way, within-subjects ANOVA to compare the mean confidence ratings for judgments

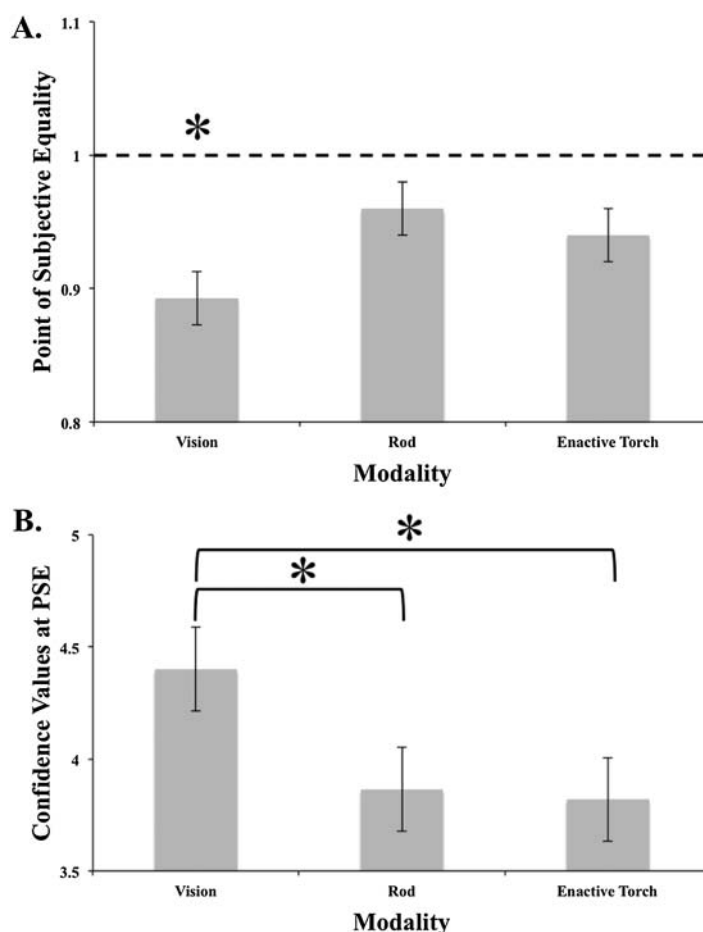


Figure 3. Mean point of subjective equality and confidence ratings for each modality. Error bars correspond to one standard error. (A) Mean point of subjective equality values for each modality, with values less than 1.0 reflecting an underestimation and values greater than 1.0 an overestimation. The point of subjective equality is scaled by actual pass-through-ability. The horizontal dashed line represents perfect accuracy in judgments; only the vision condition differed significantly from hypothetically perfect performance. (B) Mean confidence ratings for judgments made for aperture widths corresponding to the point of subjective equality for each modality, with 1 being lowest confidence and 7 being highest confidence.

made for aperture widths corresponding with the PSE for vision, rod, and Enactive Torch. There was a significant main effect of modality, $F(2, 26) = 13.46$, $p < .001$, $\eta_p^2 = .34$. Post hoc comparisons using paired-samples t -tests with Bonferroni corrections were conducted to compare confidence ratings between the vision, rod, and Enactive Torch conditions. There was a significant difference in the ratings between the vision and rod conditions, $t(26) = 4.37$, $p < .001$. There was also a significant difference in the ratings between the vision and Enactive Torch conditions, $t(26) = 5.46$, $p < .001$. However, there was not a significant difference in the ratings between the rod and Enactive Torch conditions, $t(26) = .32$, $p = .75$.

Discussion

We quantified and compared participants' performance on judgments of aperture pass-through-able affordances in order to test the hypothesis that vision would support more accurate affordance judgments than two haptic-based tools: a rod and the Enactive Torch. The data suggest, contrary to our hypothesis, that participants made perceptually equivalent judgments across all three modalities. We base this conclusion—which we offer cautiously, considering the perils of interpreting nonsignificant statistical results—on there being no significant difference among the three modalities in terms of the PSE indicating the category boundaries separating what was perceived as pass-through-able and not pass-through-able. The conclusion is further supported by additional analyses that verified the absence of a modality effect on PSE, absence of effect of order of modality on PSE, and absence of effect of time on PSE. However, when comparing judgments in terms of absolute accuracy (i.e., relative to perfect accuracy), the two haptic modalities provided judgments that did not differ from the reference, while visual judgments were significantly less accurate.

The current study is one of the few studies to directly compare affordance perception using multiple modalities (cf. Fitzpatrick et al., 1994). In addition, this study was one of the first to utilize the Enactive Torch in an experimental setting. In a preliminary study, Froese et al. (2012) found that participants could successfully navigate a simple maze using the Enactive Torch. That study did not, however, compare navigation using the Enactive Torch to navigation using another modality. Moreover, that study permitted coarse measures of task performance but did not allow the researchers to investigate the accuracy of perceptual decision-making using the Enactive Torch, *per se*.

That participants appeared to make perceptually equivalent judgments across all three modalities is consistent with the Gibsonian ecological theory of direct perception. Ordinarily, the perception of affordances facilitates successful actions, at least in part, because ambient energy arrays are rich enough in information to reliably specify action-relevant features of the environment (Favela, 2016; cf. Purves, Morgenstern, & Wojtach, 2015). The present results may indicate that both optical and mechanical (i.e., haptic) arrays contained structure to specify the pass-through-able affordance, and, moreover, that participants were sensitive to that structure. Alternatively, it remains possible that participants were employing different modes of perception across the different modality conditions used in this study. Although ecological psychology treats visual perception as direct, unmediated contact with the environment, other frameworks tend to be representational and treat vision as an indirect and inferential process. Examples of such approaches include Bayesian inference (Geisler, 2011; Kording & Wolpert, 2004), computational (Marr, 1982/2010), information theoretic (Carandini & Heeger, 2012), and reflexive (Purves et al., 2015; Purves, Wojtach, & Lotto, 2011). Full consideration of these alternatives is beyond the scope of the current work.

Despite participants' high degree of accuracy when using the rod and the Enactive Torch, participants reported that they were more confident in their perceptual judgments when utilizing visual information. This is noteworthy because there is a discrepancy between what participants were confident doing and what they were able to do: Participants were more confident with the modality with which they were least accurate (i.e., vision), and least confident in the modalities with which they were more accurate (i.e., haptic; compare Figures 3A and 3B; see also Runeson, Juslin, & Olsson, 2000). This juxtaposition of the accuracy and

confidence data is broadly consistent with the results of Froese et al. (2012), who found that while participants could navigate a maze using the Enactive Torch, participants' subjective reports indicated that the phenomenological experience was somewhat deliberative and that the Enactive Torch interface did not effortlessly facilitate the ability to act. It is also consistent with the results of Kolarik et al. (2014), who found evidence that aperture passage behavior was more cautious when using an echoic sensory-substitution device than with vision. It is plausible that more experience using the Enactive Torch than allowed in the present study or in the Froese et al. (2012) study could enhance confidence in sensory-substitution devices such as the Enactive Torch.

The results of the current study relate to previous findings in a number of noteworthy ways. First, our actual critical aperture-to-shoulder-width ratio, at a normal walking speed, of $A/S = 1.36$ is higher than Warren and Whang's (1987) $A/S = 1.30$ and Davis et al. (2010) $A/S = 1.22$. It is possible that our $A/S = 1.36$ is a more accurate reflection of the general aperture-to-shoulder-width ratio across body sizes. One reason may be that our value is based on a larger sample size ($N = 27$) than those of both Warren and Whang ($N = 10$) and Davis et al. ($N = 7$). Another reason is that our participants had a wider age range (18 to 42 years; $M = 22.31$, $SD = 6.17$ years) than for Davis et al. (24 to 28 years; $M = 25$, $SD = 1.27$ years) (Warren and Whang did not report participant ages). Accordingly, a wider age range may reflect a wider range of body types, which, in turn, may have contributed to our higher A/S value.

Second, that the Enactive Torch and rod judgments did not differ from perfect, action-scaled accuracy, while visual judgments did, adds to a finding of Burton and Cyr's (2004) concerning the relationship between experience and performance using canes. They found that visually impaired persons with experience using canes and sighted persons with no such experience had equivalent performances on gap crossability tasks utilizing rods of various lengths. Burton and Cyr noted that experience with haptic mobility assistance devices such as a cane was not necessary for "fairly high level" task performance (2004, p. 314). Although performance only among healthy sighted persons who were blindfolded was compared here, the current study takes Burton and Cyr's findings a step further: Results of the current study suggest that participants not only may demonstrate no significant difference among visual and haptic modalities, but they may be capable of also performing more accurately with haptic than visual information on tasks normally associated with visual information, such as prospectively judging whether they can pass through apertures.

One reason the rod and Enactive Torch modalities may have been more accurate than vision in the current study is because they may have involved a greater degree of deliberate information-generating or exploratory perceiver motion than did the vision condition. In the vision condition, participants opened their eyes, looked at the aperture, and then made a judgment, which was done with little movement other than postural sway. In the other two conditions, participants moved their arms, shoulders, and torso as they actively wielded the rod and Enactive Torch in order to obtain information about the aperture. This is not to suggest that participants did not produce any information-generating exploratory motion in the vision condition. Eye movements were almost certainly involved, and postural sway is always occurring. It is known that postural sway is important with regard to the accuracy of affordance judgments (e.g., Chen & Stoffregen, 2012; Stoffregen et al., 2013; Stoffregen et al., 2000; Stoffregen & Riccio, 1988). Nonetheless, in the rod and Enactive Torch conditions

participants' movements were deliberative and purposefully exploratory, perhaps because they were confronting a much less familiar task than making a visual judgment, which may have enabled more accurate judgments of the affordance boundary.

Effective sensory-substitution devices

The Enactive Torch embodies two key features that we believe are critical for effective sensory-substitution devices. First, it provides the user with information that is specific to the animal-environment system (Gibson, 1966/1983, 1979/1986), meaning that vibration amplitude maps reliably with environmental distances. Although this relation between distance and vibration amplitude is an artifactual (i.e., engineered) one rather than a natural one, the specificity of this mapping has the consequence of increased familiarity via experience, which means that the user can depend less and less on inferring the environmental layout on the basis of potentially ambiguous or unreliable information.

Second, by design the Enactive Torch requires that users engage in active exploration of the environment (Froese et al., 2012), which may enhance the user's perceptual experience over a device that promotes user passivity. Research with tactile vision substitution systems (TVSS) by Bach-y-Rita and colleagues (Bach-y-Rita et al., 1969) provided some of the earliest experimental considerations for the idea that effective perception requires active exploration. In short, the TVSS converts images captured via a video camera into a "tactile image" (Bach-y-Rita et al., 1969; Lenay et al., 2003), which is conveyed via a matrix of activators that correspond to portions of the camera lens. With little time using the TVSS, subjects were able to successfully identify basic shapes and orient themselves. Given more time (e.g., 10 hours), subjects were able to perform more complicated tasks such as identifying complex geometric shapes. The BrainPort vision device (Arnoldussen & Fletcher, 2012; Grant et al., 2016), a more recent take on TVSS-like sensory-substitution devices that applies stimulation to the user's tongue via a matrix of electrodes, has even enabled a blind user to successfully complete the highly complex task of climbing Mt. Everest (Levy, 2008). Sensory-substitution devices like the Enactive Torch, TVSS, and BrainPort draw attention to the strong possibility that movement is essential to most perceptual tasks (Lenay et al., 2003). The current study provides evidence to reinforce that claim. Moreover, this study lends preliminary support to the conclusion that haptic sensory-substitution devices like the Enactive Torch and rod (i.e., canes) support sensorimotor decision making that may be perceptually equivalent to vision, even in tasks typically assumed to require sight for making successful judgments about mobility-related affordances.

Perceptual systems as softly assembled systems

Our findings suggest equivalence of perceptual judgments among vision, haptic probing with a rod, and Enactive Torch conditions. The latter two conditions involved tools our participants had little or no previous experience utilizing and with which they had minimal familiarization and training prior to performing the experimental task. Moreover, such equivalence is further suggested by the findings that there was no effect of order of presented modality and no effect of time independent of modality. The former finding excludes explaining equivalence for those participants whose first condition was vision, which could have been said to calibrate later judgments made with the rod and Enactive Torch. The latter

finding suggests that instead of participants making more accurate judgments as the trials went on, they were able to recruit whichever modality they had available to complete the task at hand regardless of the amount of trials they had completed to that point or previous modalities utilized.

We think such findings have two important implications. First is that perception is an activity of perceptual systems, not of sensory receptors (Gibson, 1966/1983). Second, perceptual systems are highly adaptable and include the flexibility to achieve goals with the body in various ways and to incorporate artifacts or objects as functional components of perceptual systems. A way to account for such adaptability is to conceive of perceptual systems as *softly assembled systems*.

A system is “softly assembled” when it is not hardwired or preprogrammed for specific and limited ranges of outputs (Thelen & Smith, 2006). Instead, softly assembled systems are comprised of components that can be temporarily coupled in many ways. Perceptual systems can be viewed as softly assembled systems such that components of the brain, body, and environment assemble themselves into functional coordinative structures in a context-sensitive fashion (Haken, 1996; Kelso, 2016; Schiecke, Heinzel, Karch, Ploderl, & Strunk, 2016; Turvey, 2007; Wagman & Hajnal, 2016). Such softly assembled systems prioritize optimizing performance over material constitution; that is, systems adaptively utilize different combinations of elements and recruit new pathways to produce the same outcome. A consequence of viewing perceptual systems as softly assembled systems is that nonbiological elements not connected to the central nervous system (CNS) can become part of the system. Bodily, non-CNS components that contribute to perception–action are quite common in nature, for example, keratinous appendages such as insect exoskeletal hair and mammalian whiskers (Burton, 1993). Nonbodily components are common in nature as well, for example, sticks used by crows to detect and extract food from within trees (Rutz et al., 2016) and monkeys using vines for movement (Seed & Byrne, 2010).

An additional consequence of treating perceptual systems as softly assembled systems is that goal-directed task completion need not be necessarily tied to single perceptual modalities. The ability to perceive and act so as to avoid an obstacle is not only achievable via visual information detected by eyes. In fact, such an act is typically achieved via the combination of multiple overlapping perceptual modalities (cf. Stoffregen & Bardy, 2001). For example, tracking a ball that is thrown at you does not just involve the eyes. Such a task involves a combined perception that includes the visual, vestibular, and somatosensory systems (Stoffregen, Mantel, & Bardy, 2017). Consequently, system perturbations and alterations (e.g., becoming visually impaired) result in compensatory reorganization (e.g., using auditory and haptic information instead) such that the system as a whole can still complete its task.

We have claimed that understanding perceptual systems as softly assembled systems has two consequences. First is that nonbiological elements not connected to the CNS can become part of the system, and second, that goal-directed task completion need not be necessarily tied to single perceptual modalities. Following from these consequences, we think it likely that inorganic, non-CNS tools such as sensory-substitution devices can become part of softly assembled perceptual systems (e.g., Arnoldussen & Fletcher, 2012; Bach-y-Rita et al., 1969; Grant et al., 2016; Levy, 2008; and the current study). Accordingly, the current study lends preliminary support to the additional conclusion that haptic sensory-substitution devices like the Enactive Torch and rod become part of softly assembled perceptual systems that can contribute to successful judgments about mobility-related affordances.

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